

Hamiltonian Boundary Value Method HBVM(4,2)

Rank-two polynomial path with four Gauss nodes

1 Polynomial-path formulation

Let $(c_i, b_i)_{i=1}^4$ be four-point Gauss–Legendre quadrature on $[0, 1]$. For the first two orthonormal shifted Legendre polynomials,

$$P_0(c) = 1, \quad P_1(c) = \sqrt{3}(2c - 1),$$

define

$$I_{i0} = c_i, \quad I_{i1} = \sqrt{3}(c_i^2 - c_i).$$

HBVM(4,2) solves for two vector coefficients γ_0, γ_1 :

$$Y_i = y_n + h \sum_{j=0}^1 I_{ij} \gamma_j, \quad (1)$$

$$R_j(\gamma) = \gamma_j - \sum_{i=1}^4 b_i P_j(c_i) f(t_n + c_i h, Y_i) = 0. \quad (2)$$

The accepted state is

$$y_{n+1} = y_n + h \sum_{i=1}^4 b_i f(t_n + c_i h, Y_i).$$

Equivalently, the Runge–Kutta matrix is $A = \mathcal{I}_2 \mathcal{P}_2^T \Omega_q$ and has rank two. The nonlinear unknown therefore has twice the state dimension, although four stages evaluate the line integral.

2 Order and energy property

With polynomial degree $s = 2$, the method has classical order $2s = 4$. Four-point Gauss quadrature has degree seven. For an autonomous polynomial Hamiltonian of degree ν , exact polynomial energy conservation follows when $k \geq \nu s/2$; hence HBVM(4,2) covers degrees through four. For a general smooth Hamiltonian the quadrature error makes energy conservation approximate, with an error that decreases rapidly with the step and quadrature accuracy.

HBVM(k, s) is generally not symplectic when $k > s$. Thus HBVM(4,2) must not be labeled symplectic by construction; map-Jacobian diagnostics measure its canonical defect independently of its energy behavior.

3 Nonlinear configurations

Newton acts on the reduced residual (R_0, R_1) . The public `jacobian_method` values are `analytic`, `finite_difference`, and `auto`. Analytic mode assembles exact guiding-centre particle blocks at the four stages; finite-difference mode differentiates the stage fields; auto selects the available capability. A backtracking correction is used only when the full Newton update increases the residual. The stopping scale is

$$\varepsilon_{\text{abs}} + \varepsilon_{\text{rel}} \max(1, \|y_n\|_\infty).$$

With `track_energy=True`, the implementation stores the physical Hamiltonian and its drift at saved times. For explicitly time-dependent systems this is a diagnostic, not an invariant claim.

4 Implementation correspondence

Coefficients, residuals, Newton assembly, and integration are in `src/simulation/methods/hbvm/order4.py`. The simulation document describes required Hamiltonian capabilities, work counters, fixed-grid sampling, and evaluation studies; shared physical dynamics are documented outside the model tree.